Yashvardhan Shashikant Borude

Software Engineer | Systems, AI/ML & Competitive Programmer

yashvardhanborude7@gmail.com | +91-7058230773 | Pune, Maharashtra, India | linkedin.com/in/yashvardhan-borude | github.com/Yashvardhan-4

Competitive Profiles: LeetCode (315+ Solved) | Codeforces (Pupil, 1227) | CodeChef (2-Star, 1434) | HackerRank (5-Star C++)

Professional Summary

Computer Engineering undergraduate (**CGPA: 8.86/10**, PCCOE Pune) with proven production engineering experience designing and deploying scalable full-stack web applications, desktop systems, and deep learning architectures. **Global Top 1,500 in Google BigCode 2026** with **600+ algorithmic problems solved** across LeetCode (315+), Codeforces, and CodeChef. Specialized in high-throughput network concurrency, low-level OS file descriptors, Selective State-Space Models (SSMs), and PostgreSQL ACID systems.

Technical Skills

Languages: C++, Python, TypeScript, JavaScript (ES2023+), SQL (PostgreSQL), GLSL, HTML5, CSS3
AI/ML & Scientific: PyTorch, Mamba (Selective SSMs), Multi-Modal Signal Processing, SciPy, Scikit-Learn, NumPy, Pandas
Frontend & Web: React 19, Next.js 16 (App Router), Three.js, React Three Fiber, Tailwind CSS v4, WebGL, Canvas API, PWA
Backend & Systems: Node.js (ESM), FastAPI, Supabase, PostgreSQL 15+, PL/pgSQL Stored Procedures (RPCs), WebSockets, TanStack Query
Developer Tools: Electron, Chrome Extensions (MV3), POSIX File Descriptors, Linux/Bash, Git, GitHub Actions, Postman

Work Experience

Shivkush Hi-Tech Nursery — **Ahilyanagar, India**
Full Stack Software Engineer (Paid Freelance Contract) — *Dec 2025 – May 2026*

- Architected an enterprise ERP (**SNMS**) in **Next.js 16, TypeScript, and Supabase PostgreSQL** (18+ migrations, 18.5K+ LOC); eliminated 100% of overselling race conditions via atomic stored procedures with **FOR UPDATE pessimistic row locking**.
- Engineered an append-only financial ledger and dynamic views (**vw_daily_cashbook**), achieving **100% cash-on-hand audit accuracy** across multi-batch seasonal dispatches by decoupling payment streams from mutable order balances.
- Built an offline-first bilingual (English/Marathi) web platform featuring a real-time **Three.js** 3D plant canvas; slashed 3D asset transfer sizes by **83.9%** (26.1MB to 4.19MB) using **Google Draco compression** to achieve 60 FPS mobile rendering.

Projects

CardioMamba-EM – Physics-Guided Linear-Time Multimodal SSM | *PyTorch, Mamba, Python, FastAPI, React* *Oct 2026*

- Designed an end-to-end continuous-time **Selective State-Space Model (BiMamba)** for synchronized 1D ECG/PCG biosignals (500 Hz), achieving strict $\mathcal{O}(L)$ linear complexity and surpassing published literature on **PhysioNet CinC 2016** (**98.49% Acc, 99.56% Sens**).
- Engineered Coarse-to-Fine Cross-Modal Lag Attention discovering significant electromechanical delay dispersion ($p = 3.39 \times 10^{-21}$); distilled model into **MicroCardioMamba (112K params, 448 KB, ∼160ms latency)** deployed with an interactive telemetry console.

TurboIDM – Multi-Stream Download Accelerator | *Node.js, Electron, React 19, MV3, POSIX fs* | GitHub *Jun 2026*

- Architected a cross-platform download manager accelerating throughput by **up to 8x** via 32-stream **HTTP Range multiplexing** and a dynamic **work-stealing algorithm** that continuously bisects straggler byte ranges to eliminate tail latency.
- Engineered zero-copy disk writes using **POSIX file descriptors** and sparse pre-allocation (**ftruncate**), cutting merge latency to **0.00s**; built a **Chrome MV3** extension capturing session cookies via a local **REST/WebSocket daemon**.

Inkless – Local-First Offline Journal & WebGL Shader Canvas | *Next.js 16, React 19, Dexie.js, Supabase, GLSL* *Sep 2026*

- Architected a local-first platform on **Next.js 16, React 19, and IndexedDB (Dexie.js)**, achieving **<5ms local write latency** (50x faster than cloud APIs) and **0% data loss** during network disruptions via an atomic transactional outbox queue.
- Engineered a background sync worker with **Block-Level Last-Write-Wins (LWW) conflict resolution** in PostgreSQL, and built a 60/120 FPS 3D canvas with **Three.js and custom GLSL noise shaders** streaming uniforms via a mutable ref pipeline.

Education

Pimpri Chinchwad College of Engineering (PCCOE) — **Pune, India**
Bachelor of Technology in Computer Engineering | ***CGPA: 8.86 / 10.0*** *(Till Semester 4)* — *Expected May 2028*

- **Academic Standing:** Third Year (Semester 5) | **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks, Machine Learning, Generative AI.

Certifications

NPTEL Elite + Silver Certificate – IIT Madras / Kharagpur — **Pune, India**
Algorithmic Graph Theory and Data Structures (12-Week Rigorous IIT Curriculum) — *Jan 2025 – Apr 2025*

- Achieved **Score: 77/100 (Elite + Silver, Top Tier)**; mastered graph traversals, Dijkstra's, Bellman-Ford, Ford-Fulkerson maximum flow, bipartite matching, Eulerian/Hamiltonian paths, and NP-completeness.

Achievements & Honors

- **Google BigCode 2026:** Ranked among the **Global Top 1,500 participants** worldwide in competitive algorithmic engineering.
- **Competitive Programming Excellence:** Solved **600+ algorithmic problems**; **LeetCode 315+ solved**, **CodeChef 2-Star** (Peak: 1434), **Codeforces Pupil** (Max: 1227, 200+ solved), and **HackerRank 5-Star in C++** (Trees, Graphs, DP).
- **Paid Commercial Software Contract:** Commissioned by Shivkush Hi-Tech Nursery as lead engineer to architect both their enterprise ERP (SNMS) and client-facing digital platform (2025–2026).
- **Smart India Hackathon (SIH 2024):** Successfully qualified and delivered rapid-response disaster preparedness MVP within a 24-hour national competition timeline.